ThreadForge

Connectors, Access, Repositories & The 20-Desk Industrial Park Suite

A comprehensive six-part lecture and architectural reference on how ThreadForge connects to local and external AI models, enforces least-privilege access and maker-checker governance, secures dual-store releases, delivers all 20 deterministic business and industrial desks offline on `127.0.0.1`, and evolves in continuous multi-model isolation.

01 Part 1 · AI, Connectors & 14-Model Isolated Co-Engineering (Slide + Notes)

02 Part 2 · Access, Named Tokens & Maker-Checker Governance (Slide + Notes)

03 Part 3 · How It Connects: End-to-End 127.0.0.1 Request Map (Slide + Notes)

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06 Part 6 · Cost-Benefit vs Legacy ERP & Pioneer Engineering (Slide + Notes)

07 Assessment 1 · Comprehensive Knowledge Check (16 Scored Questions)

08 Assessment 2 · Four Practical Labs & Enterprise Design Proposal

09 Assessment 3 · Marking Rubric & Evidence Grading Bands

10 Teacher & Architect Answer Key (16 Knowledge Check Answers + Lab Solutions)

Learning & Engineering Outcomes

- Explain where an AI model runs (local Ollama on `127.0.0.1:11434` vs provider accounts) and how 14 AI model contexts co-engineer ThreadForge in continuous isolation.
- Issue, audit, and revoke least-privilege organization tokens and enforce two-stage Director + Workspace Owner executive approvals.
- Map all **20 desks and toolkits** across Essentials ($49), Professional ($149), Industrial ($299), Industrial Park ($499), and Custom ($19 + $15/desk).
- Quantify 5-year TCO savings against SAP S/4HANA, Oracle NetSuite, SAP Business One, and Odoo while preserving 100% offline data sovereignty.

The Pioneer-Class Principle

"The agent stack proposes; the human disposes." ThreadForge is trained and verified across **Argon, Opus, Codex, Gemma, Beam, DeepSeek Flash, Gemini Flash, GPT Sol, GPT Terra, Qwen 31B, Cursor, Grok Bot, SuperGrok, and OpenAI** in sealed isolation with `639` Python unit tests, `28` Node tests, and `52` Windows native checks running simultaneously. Every calculation is executed by deterministic Python code, and every software licence is verified offline via Ed25519 cryptography.

`Official Stores: Microsoft Store (9PJTNXTTX5CR) · Google Play (com.methodology.threadforge)`

THREADFORGE 10.18 INDUSTRIAL PARK · LECTURE · PART 1

AI, Connectors & Multi-Model Charter

ThreadForge

You choose where the model runs. Nothing leaves your PC until the owner enables, tests, and saves it — backed by 14-model isolated co-engineering.

1 Where the model runs

- **Ollama on this device:** free local model at `127.0.0.1:11434`. No API key, no per-token bill, zero data leaves the PC.
- **Ollama on LAN / private host:** over HTTPS for a dedicated GPU workstation on your own network.
- **Your own provider account:** OpenAI, Anthropic, Google Gemini, or any OpenAI-compatible endpoint. Billed directly by the provider with zero ThreadForge markup.
- **Builder + reviewer pair:** optional second model gives an independent second opinion before any code or document review completes.

2 MCP servers & plugins

- Listed in owner-controlled files on the server: an MCP registry and one folder per plugin with a `threadforge-plugin.json` manifest.
- **Everything starts disabled.** The workspace owner enables each connector and runs a live connection check.
- Only the explicit `allowed_tools` declared in the owner's file are exposed to an agent.
- Tool outputs are treated strictly as **untrusted data**, never as executable instructions.
- **Deterministic math rule:** all financial, tariff, CBM, and actuarial arithmetic is computed by ThreadForge's Python engines, never by an LLM.

3 Multi-Model Isolated Charter

- Co-engineered & stress-tested in continuous isolation across **14 frontier model & agent contexts**:

`Argon` `Opus` `Codex` `Gemma` `Beam` `DeepSeek Flash` `Gemini Flash` `GPT Sol` `GPT Terra` `Qwen 31B` `Cursor` `Grok Bot` `SuperGrok` `OpenAI`

- **Simultaneous test gates:** every isolated run must pass `639` Python unit tests, `28` Node tests, and `52` Windows native checks.
- **Pioneer-Class Principle:** *"The agent stack proposes; the human disposes."* Engineering agents open draft PRs only and never auto-merge.

1 · CHOOSE
Ollama on `127.0.0.1`, LAN GPU host, or your own provider account

2 · CONFIGURE
Endpoint + model name (e.g. `qwen2.5-coder:7b`) · key only for external providers

3 · TEST CONNECTION
Single deterministic probe verifies the model responds before saving

4 · SEAL & SAVE
Secrets stay in owner-protected server files and are never echoed to browsers

Local AI with Ollama	Provider accounts, plugins, MCP	Engineering agents + Sandbox	Custom Single-Desk Option
Professional ($149) and above	Industrial ($299) and above	Industrial ($299) and above	Pick only the desks you need: $19 + $15/desk

Source: ThreadForge 10.18.0 · business_connectors.py, licence.py, PIONEER-CLASS-PRINCIPLE.md

threadforgeofficial.com

ThreadForge

AI, Connectors & Multi-Model Co-Engineering

Key Architectural Principles

- **Local first on 127.0.0.1:** Ollama on the same PC (`http://127.0.0.1:11434/v1/chat/completions`) needs no API key, costs $0 per answer, and sends zero bytes outside the machine.
- **Zero provider markup:** When an owner connects OpenAI, Anthropic, or Gemini, billing is direct with the provider.
- **Off by default:** MCP servers and plugins (`threadforge-plugin.json`) do nothing until the workspace owner enables and tests each one.
- **Untrusted tool results:** Tool outputs are quarantined as plain data, never interpreted as agent instructions (preventing prompt injection).
- **Deterministic math boundary:** All 50+ TradeMath calculators, ERP totals, and PWI actuarial formulas run in deterministic Python/JS — never inside an LLM.

Classroom & Team Demonstration

- **Step 1:** Open *AI & models* desk. Select **Ollama on this device**.
- **Step 2:** Enter `http://127.0.0.1:11434/v1/chat/completions`, model `qwen2.5-coder:7b`, leave API Key empty.
- **Step 3:** Click **Test the connection**, verify deterministic response, then click **Save**.
- **Step 4:** Demonstrate input validation: typing a conversational sentence into the Model field triggers a clear validation warning rather than a silent failure.

Common Misconceptions

- *"The AI calculates customs duty and CBM."* False: ThreadForge's deterministic calculators compute all figures; AI only reviews text or drafts plans.
- *"An API key is required for local AI."* False: Ollama on loopback requires no key.

14-Model Isolated Co-Engineering

Under the ThreadForge Academy Charter, the suite is continuously trained, cross-audited, and evolved in isolated sandboxes across:

`Argon · Opus · Codex · Gemma · Beam · DeepSeek Flash · Gemini Flash · GPT Sol · GPT Terra · Qwen 31B · Cursor · Grok Bot · SuperGrok · OpenAI`

- **Simultaneous test runs:** Each isolated agent branch must pass `639` Python unit tests, `28` Node tests, and `52` Windows checks before human review.
- **Discussion Question:** Why does running a second independent "reviewer" model alongside deterministic unit tests catch defects that a single model misses?

THREADFORGE 10.18 INDUSTRIAL PARK · LECTURE · PART 2

Access, Tokens & Governance

ThreadForge

One workspace owner, up to 100 isolated organization SQLite databases, named least-privilege tokens, and maker-checker executive approvals.

1 Who holds what

WORKSPACE OWNER
Holds the master workspace token. Controls engineering, host & provider settings, Ed25519 licence activation, backup, restore, and final Stage-2 Executive Approvals.

Isolated Organizations (up to 100)
Each organization is a separate SQLite file on disk. Created by the owner only — zero public sign-up, zero cross-tenant leakage.

Named Role Tokens & Director Sign-Off
Bound to a single organization and role. Submitters cannot approve their own executive proposals (maker-checker separation).

2 Four organization roles

ROLE	PERMITTED ACTIONS
Admin	All organization desks, HR attendance review, executive proposal submission, and organization tokens/invitations.
Operator	Inventory, orders, imports, sources, campaigns, projects, capital projects, R&D/3PL/mfg toolkits, PC Care, due diligence.
Finance	Payments, invoices, contacts, trade, rate & policy log, treasury/audit checklists, and finance executive proposals.
Viewer	Read-only across records. May run all 50+ TradeMath calculators, view handouts, and complete Mentor learning paths.

Every signed-in role can read organization records, download lecture handouts, and consult the Mentor.

3 Golden security rules

- **One person, one named token.** Never share an admin token across staff; named tokens make the audit ledger accountable.
- **Shown once at creation.** The server stores only a keyed cryptographic digest, so a lost token is replaced, never recovered.
- **Maker-Checker Executive Approvals:** Finance/Admin submits proposal → separate Director records Stage-1 decision → Workspace Owner records final Stage-2 decision.
- **Isolated engineering credentials:** worker, sandbox, model key, and GitHub token are strictly separated; credential reuse is refused.
- **Permanent keys stay air-gapped:** Ed25519 licence signing and Android release keys never reside on shared machines.

1 · ISSUE
Name + role + expiry of 1–365 days (or 7-day one-time invite code)

2 · SHOWN ONCE
Only a keyed digest is stored in the organization SQLite database

3 · USE
12-hour browser session cookie paired with mandatory `X-CSRF-Token`

4 · REVOKE ANYTIME
Takes effect on the very next HTTP request with zero server restart

Invitations
One-time code, valid 7 days

Hard Limits
100 active tokens & 50 invitations / org

Access & Policy Ledger
Every token & approval event is timestamped

Executive Approvals Desk
Two-stage director + owner governance (Industrial Park)

Source: ThreadForge 10.18.0 · tenancy.py, server.py, TOKEN-ACCESS-POLICY.md, INDUSTRIAL-PARK.md

Access, Tokens & Maker-Checker Governance

Key Access Principles

- **Strict multi-organization isolation:** One workspace supports up to 100 organizations, each stored in its own separate SQLite file on disk.
- **Least privilege by role:** Every named token is bound to one organization and one role (`admin` , `operator` , `finance` , or `viewer`).
- **Shown once, stored as digest:** The raw token is displayed once upon creation; the database stores only a keyed cryptographic digest.
- **1–365 day expiry & instant revocation:** Revoking a token invalidates access on the very next HTTP request without restarting the server.
- **Separate engineering credentials:** Worker, sandbox, model, and GitHub tokens are distinct; reusing one across boundaries is refused.

Live Demonstration

- **1. Issue viewer token:** In *People & access*, issue a token named `Trainee · viewer` valid for 1 day.
- **2. Verify read-only guard:** Sign in with that token in a private browser window and attempt to add a stock item — the gateway refuses with *"This organization credential is read-only."*
- **3. Run TradeMath:** Execute a CBM or landed-cost calculation — it succeeds because calculation mutates no records.
- **4. Maker-Checker Executive Approval:** Submit a capex proposal as `finance` , record Stage-1 review as a separate `director` , and finalize Stage-2 approval as the `workspace owner` .

Discussion Questions

- **1. Accountability:** Why is sharing a single admin token across five employees far more dangerous than issuing five named tokens?
- **2. Segregation of duties:** Finance can record payments and invoices, while Operators manage stock movements and warehouse receipts. What classic internal fraud does this separation prevent?
- **3. Maker-Checker Governance:** Why does the Industrial Park *Executive Approvals* desk forbid a proposal submitter from recording the Director or Owner decision on their own proposal?
- **4. Digest storage:** If an attacker copies an organization SQLite file, why can they still not read the plaintext access tokens?

THREADFORGE 10.18 INDUSTRIAL PARK · LECTURE · PART 3

How It Connects

Ten words, one deterministic map: who calls what, with which cryptographic proof, and where every component runs offline on `127.0.0.1`.

ThreadForge



ROLE

What a credential may change inside one organization.

`operator — stock, orders, toolkits; not payments`

ROUTE

One address on the server for one deterministic action.

`POST /api/suite/trade/calculate`

AUTH

How the server verifies who is calling on every request.

`Authorization: Bearer <token> or cookie + X-CSRF-Token`

MCP

Model Context Protocol: standard way to give agents tools.

`"docs": transport http, allowed_tools ["search"]`

API

The complete set of HTTP routes that clients call.

`GET /api/suite/licence → tier, features, days left`

TOKEN

A secret that proves identity & role. Shown once.

`"Ayesha · finance", 90 days, one organization`

SERVER

The local Python gateway answering routes on loopback.

`tf-agent serve → 127.0.0.1:8787`

PROVIDER

The runtime executing the AI model weights.

`Ollama 127.0.0.1:11434 or your Anthropic/OpenAI key`

RUNNER

Executes code and test suites isolated from the server.

`run-worker + broker :8790 (Docker) + GitHub CI`

HOST

The physical machine and address where the server lives.

`Your PC (127.0.0.1) or your private server via tunnel`

Source: ThreadForge 10.18.0 · server.py, tenancy.py, licence.py, business_connectors.py

threadforgeofficial.com

How It Connects: End-to-End Request Trace

Key Connection Rules

- **Every arrow carries proof:** No internal or external hop trusts network location alone; every request presents its own credential.
- **Browser session + CSRF:** Signing in exchanges a named token for a 12-hour session cookie (43,200s), and every state-changing request requires a matching `X-CSRF-Token` header.
- **Three gateway gates:** Every API call passes through (1) Auth verification, (2) Route & Ed25519 Licence Tier check, and (3) Organization Role check before touching SQLite.
- **Inbound worker lease:** The Docker test runner calls into `127.0.0.1:8790` with a dedicated worker token; the gateway never opens arbitrary inbound shell ports.

Guided Trace Exercise

- **Scenario A (Allowed):** An `operator` saves a stock movement in the browser.
  1. *Auth:* 12h session cookie + `X-CSRF-Token` verified.
  2. *Route & Tier:* `POST /api/suite/items` requires `business` feature (unlocked in Essentials+).
  3. *Role:* `operator` is permitted to modify inventory.
  4. *Database:* Organization SQLite file updated and ledger entry written.
- **Scenario B (Blocked at Role):** Repeat the exact request signed in as `finance` — stopped at Gate 3 with *"outside the credential's organization role"*.

Discussion Questions

- **1. Loopback Sovereignty:** Why does binding the server to `127.0.0.1:8787` by default protect a laptop user on public airport or café Wi-Fi?
- **2. CSRF Defense:** Why is a session cookie alone insufficient for `POST` requests in a browser, and how does `X-CSRF-Token` stop cross-site request forgery?
- **3. Container Isolation:** Why do engineering agent test runs execute inside an isolated Docker container rather than directly on the host operating system?
- **4. Offline Continuity:** Trace which arrows on the map require internet access when using local Ollama. (Answer: None.)

THREADFORGE 10.18 INDUSTRIAL PARK · LECTURE · PART 4

Repositories, Stores & Release Path

ThreadForge

Two isolated repositories, hardware-protected signing environments, Dual-OS global store distribution, and zero-trust CI verification.

1 Private: ThreadForge-Inc/SO

- **Complete source code,** 20 suite desks, Chaquopy Android bridge, Electron/MSIX desktop shell, and 639 Python + 28 Node tests.
- **GitHub-hosted CI matrix:** Windows 11, Android API 35, macOS, MSIX packaging, and VS Code extension verification.
- **Cryptographic provenance:** every release artifact is pinned to exact CI run IDs and verified by SHA-256 digest before publication.
- **Clean git topology:** zero stale worktrees, zero unreviewed auto-merges, and strict branch protection on `main`.

2 Public: threadforge-website

- **Live global site:** `threadforgeofficial.com` served over HTTPS with strict Content-Security-Policy (0 inline scripts/styles).
- **Automated site auditor:** `check_site.py` verifies 22 HTML pages, 13 CSS stylesheets, local links, store URLs, and CSP compliance on every push.
- **Dual-OS Store Parity:** Microsoft Store (`9PJTNXTTX5CR`) and Google Play (`com.methodology.threadforge`) + direct signed installers.
- **Source code never goes here:** source archives and internal wheels remain strictly inside private `ThreadForge-Inc/SO`.

3 Who may do what

ACTOR	ACCESS & CRYPTOGRAPHIC BOUNDARY
Owner account	Admin on both repos. Approves merges to `main` and unlocks `android-release` signing.
Pull-request CI	Read-only permissions. Signs PR test builds with an ephemeral throwaway key only.
Release signing	Protected `android-release` environment holds permanent upload key (`95387cee...`).
Play Edition build	`-PtfEdition=google_play` strips external purchase links & uses Google Play Billing 7.1.1.
Engineering agents	Scoped fine-grained token in protected file; draft PRs only on allowlisted repositories.

MILESTONE 1 · CORE & ACADEMY
v10.0–10.15: Local SQLite ERP, 50+ TradeMath calculators, Due Diligence, Mentor, 16-Work Academy & 5 PWI Labs

MILESTONE 2 · STORE EDITIONS
v10.16.1: Microsoft Store (`9PJTNXTTX5CR`) & Google Play Store (`com.methodology.threadforge` , Play Billing 7.1.1)

MILESTONE 3 · ED25519 & HANDOUTS
v10.17: Offline Ed25519 licence keys, pay-the-difference upgrades, Custom $19+$15/desk tier, in-app PDF handouts

MILESTONE 4 · INDUSTRIAL PARK
v10.18.0: Complete 20-Desk Suite — HR Attendance, Executive Approvals, Team Toolkits, Learning Paths, Image Workshop

639 Python + 28 Node Tests
100% passing across backend, web & handouts

0 CSP / Site Errors
Verified across all 22 pages by check_site.py

Signed AAB & APK Ready
10.16.1 (101601) & 10.18.0 (101800) signed with 95387cee...

Sovereign Offline Operation
Zero cloud dependency after installation

Source: ThreadForge 10.18.0 · release workflows, check_site.py, BillingManager.java, TOKEN-ACCESS-POLICY.md

threadforgeofficial.com

Repositories, Dual Stores & Milestone Journey

Two-Repository Architecture

- **Private Source (`ThreadForge-Inc/SO`):** Houses all Python backend code, 20 web desks, Chaquopy Android studio project, Electron/MSIX desktop client, VS Code extension, and 719+ automated tests.
- **Public Website (`ThreadForge-Inc/threadforge-website`):** Powers `threadforgeofficial.com` via GitHub Pages. Contains static HTML/CSS/JS, Academy monographs, lecture handouts, and verified binary release downloads — never source code zips.
- **Automated Site Gate (`check_site.py`):** Audits all 22 public HTML pages and 13 CSS files for broken links, duplicate IDs, third-party trackers, and strict CSP compliance.

Dual-OS Store Compliance & Signing

- **Microsoft Store (`9PJTNXTTX5CR`):** Native Windows desktop installer and MSIX package across all 4 tiers.
- **Google Play Store (`com.methodology.threadforge`):** Built with `-PtfEdition=google_play`, integrating native Google Play Billing `7.1.1` (`BillingManager.java`) and stripping external purchase links to satisfy Play Families & Payments policy.
- **Hardware/Environment Key Isolation:** Pull-request CI builds use an ephemeral throwaway key. Only owner-approved runs on `main` inside the protected `android-release` environment can access the permanent upload key (SHA-256 `95387cee...`).

The Milestone Journey (10.0 → 10.18.0)

- **v10.0–10.15:** Core SQLite business suite, 50+ TradeMath calculators, Supplier Due Diligence, Mentor, and 16-Work Online Academy with 5 PWI Practice Labs.
- **v10.16.1:** Official Microsoft Store & Google Play Store Dual-OS release parity and 7 HD white-label themes.
- **v10.17.0–10.17.1:** Offline Ed25519 licence verification, 14-day full trial, Custom $19+$15/desk tier, and in-app PDF lecture handouts.
- **v10.18.0 Industrial Park:** Complete 20-capability enterprise suite adding HR & Attendance, Executive Approvals, Team Toolkits, Mentor Learning Paths, and Local Image Workshop.

THREADFORGE 10.18 INDUSTRIAL PARK · LECTURE · PART 5

20 Suite Desks & Industrial Park Tiers

ThreadForge

Every desk and toolkit is built into the same offline-first codebase and unlocked by a signed Ed25519 licence key — paid once, never rented.

E Essentials · $49 once

6 CORE DESKS INCLUDED

- **1. Business Workspace (`business`):** Stock, orders, invoices, payments, contacts, sources, campaigns, rate & policy log.
- **2. Projects (`projects`):** Tasks, milestones, owners, and delivery tracking.
- **3. Starter TradeMath (`trade`):** Core CBM, container load, and landed-cost calculators.
- **4. Supplier Due Diligence (`diligence`):** 8-source verification checklist & audit trail.
- **5. The Mentor (`mentor`):** Plain-language operational guidance & next steps.
- **6. Custom App Branding (`white_label`):** 7 HD themes, logo, company name & invoice header.

P Professional · $149 once

EVERYTHING IN ESSENTIALS + 6 DESKS (12 TOTAL)

- **7. Specialist TradeMath (`trade_specialist`):** All 50+ calculators (Incoterms 2020, LC, FX, polymers, GSM, PCB, safety).
- **8. PC Care Diagnostics (`pc_care`):** Read-only Windows health inspection & native DISM/SFC/Defender dispatch.
- **9. App Builder & Studio (`app_builder`):** Turn plain-word specs into working local web apps.
- **10. Office Files & Doc Studio (`office_files`):** Formula-safe CSV/Excel/PDF import-export, certificates, ID cards, stamps.
- **11. AI Document Review (`document_review`):** Second-pass review of calculations & trade docs.
- **12. Local AI with Ollama (`local_ai`):** Private on-device LLM at `127.0.0.1:11434` .

I Industrial · $299 once

EVERYTHING IN PROFESSIONAL + 4 DESKS (16 TOTAL)

- **13. AI Connectors & MCP (`ai_connectors`):** Owner-approved OpenAI/Anthropic/Gemini keys, plugins, and MCP servers.
- **14. Engineering Agents (`engineering_agents`):** Builder + reviewer agents with Docker sandbox verification & draft PRs.
- **15. Capital Projects (`capital_projects`):** Multi-stage capex budgets, variations, and milestone gates.
- **16. HR & Attendance (`hr_payroll`):** Employee roster, cadres, attendance CSV preview/import, admin approval & totals export.

IP Industrial Park · $499

ALL 20 CAPABILITIES + SOURCE ENTITLEMENT

- **17. Executive Approvals (`executive_approvals`):** Two-stage Director + Workspace Owner maker-checker register.
- **18. Team Operational Toolkits (`team_toolkits`):** R&D, manufacturing, 3PL transport & finance/audit checklists + custom 40-task workbooks.
- **19. Mentor Learning Paths (`learning_paths`):** 4 self-paced lessons with 80%-pass scored assessments.
- **20. Local Still-Image Workshop (`image_workshop`):** Browser crop, rotate, resize & colour studio with brand library save + **Audit, Finance & Treasury (`finance_treasury`).**

14-Day Full Trial Built In
Every new install unlocks all 20 Industrial Park desks for 14 days offline

Custom Tier: $19 + $15 / desk
Single-desk entry from $34 once; never pay for desks you don't use

Pay-Only-The-Difference Upgrades
Upgrade Essentials → Professional for $100, or Industrial → Park for $200

Unlimited Users & 100 Orgs
No per-seat tax, no annual maintenance fee, no cloud lock-in

Source: ThreadForge 10.18.0 · src/threadforge/licence.py (FEATURES, TIER_FEATURES, TIER_PRICES_USD), INDUSTRIAL-PARK.md

threadforgeofficial.com

All 20 Suite Desks & Industrial Park Capabilities

Complete 20-Capability Matrix (`src/threadforge/licence.py`)

TIER & ONE-TIME PRICE	FEATURE KEYS & OPERATIONAL PROBLEM SOLVED
Essentials `$49 once`	1. `business` : Stock, orders, invoices, payments, contacts, sources. local campaigns, policy log. 2. `projects` : Project boards, milestones, task owners & deadlines. 3. `trade` : Starter CBM, container loading & landed-cost calculations. 4. `diligence` : 8-source supplier due diligence workflow & evidence log. 5. `mentor` : Context-aware plain-language operational guidance. 6. `white_label` : 7 HD brand themes, logo upload & branded invoice headers.
Professional `$149 once`	7. `trade_specialist` : All 50+ TradeMath calculators (Incoterms 2020, LC, FX, polymers, GSM, PCB, safety). 8. `pc_care` : Read-only WMI/SMART diagnostics & consented native SFC/DISM/Defender dispatch. 9. `app_builder` : App Builder & Studio for generating local business web apps. 10. `office_files` : Formula-safe CSV/Excel/PDF interchange + Document Studio (certificates, cards, stamps). 11. `document_review` : AI-assisted review of trade documents and calculations. 12. `local_ai` : On-device Ollama integration at `127.0.0.1:11434`.
Industrial `$299 once`	13. `ai_connectors` : External AI providers, plugins & allowlisted MCP servers. 14. `engineering_agents` : Builder + reviewer code agents with Docker sandbox testing. 15. `capital_projects` : Multi-stage capital expenditure budgets. variations & gates. 16. `hr_payroll` : Employee roster, cadres, attendance CSV import, admin review & approved workdays export.
Industrial Park `$499 once`	17. `executive_approvals` : Maker-checker register (Submitter → Director → Workspace Owner). 18. `team_toolkits` : R&D, manufacturing, 3PL & finance/audit checklists + 40-task custom workbooks.

Honest Delivery Boundaries (`INDUSTRIAL-PARK.md`)

- **HR & Attendance:** Tracks employee numbers, cadres, and approved attendance totals; purposefully avoids collecting government IDs, bank details, or medical records.
- **Executive Approvals:** Records auditable two-stage governance decisions in USD cents with safe evidence references; never auto-executes bank transfers or alters ledger balances behind the scenes.
- **Local Still-Image Workshop:** Processes PNG/JPEG/WebP up to 32 MiB and 16 MP entirely in the browser; saving to the organization brand library (up to 1 MiB) is an explicit separate action.
- **Private Campaign Capture:** Outbound SMTP mass-mailing is disabled by design so workspaces never become spam relays.

Offline Ed25519 Licensing & Custom Tier

- **One binary, instant unlock:** All 20 desks ship in the standard installer. Entering an Ed25519-signed key unlocks the purchased tier offline with zero phone-home servers.
- **Custom Tier ($19 + $15/desk):** A warehouse needing only Business + Specialist TradeMath pays `$19 + $30 = $49`; a studio needing only Image Workshop pays `$19 + $15 = $34` once.
- **Pay-only-the-difference upgrades:** Every dollar spent on a lower tier credits 100% toward higher tiers.

THREADFORGE 10.18 INDUSTRIAL PARK · LECTURE · PART 6

Cost-Benefit & Pioneer Engineering

ThreadForge

Why one-time deterministic software on `127.0.0.1` beats recurring cloud ERP tenancy on TCO, speed, privacy, and resilience.

1 10-User Enterprise TCO: ThreadForge vs Legacy Competitors

PLATFORM / TARGET	BILLING MODEL	YEAR 1 (10 USERS)	5-YEAR TCO	DATA & HOSTING
SAP S/4HANA Cloud	Per-user / mo subscription	**$32,400 – $72,000+**	**$162,000 – $360,000+**	Mandatory vendor cloud
Oracle NetSuite	Base platform + per-seat/mo	**$24,000 – $60,000**	**$120,000 – $300,000**	Oracle cloud tenancy only
SAP Business One	Named user + partner licence	**$13,200 – $26,400**	**$66,000 – $132,000**	Partner server / cloud
Odoo Standard / Custom	Per-user / mo + app add-ons	**$3,732 – $8,900+**	**$18,660 – $44,500+**	Metered cloud or self-host
ThreadForge Custom (1 Desk)	**One-time ($19 + $15/desk)**	**$34 once ($0/mo)**	**$34 total**	**100% Local 127.0.0.1**
ThreadForge Essentials / Pro	**One-time perpetual tier**	**$49 / $149 once**	**$49 / $149 total**	**Unlimited users · 100 orgs**
ThreadForge Industrial Park	**One-time full 20-desk suite**	**$499 once ($0/mo)**	**$499 total (99% saved)**	**Full Suite + Source Code**

2 Deterministic Engineering

- **Zero cloud latency or outages:** every desk runs against local SQLite on `127.0.0.1` in milliseconds — even when submarine cables or ISP links fail.
- **Formula-Safe ERP Interchange:** `_csv_cell` automatically escapes leading `=`, `+`, `-`, `@` to block spreadsheet formula injection across SAP/NetSuite/Excel exports.
- **Closed-Form TradeMath:** 50+ calculators execute exact decimal & actuarial equations rather than probabilistic LLM guesses.
- **Cryptographic Licence Integrity:** Ed25519 offline verification unlocks tiers instantaneously without calling a licensing server.

3 14-Model Evolution

Trained and evolved in continuous isolation under the ThreadForge Academy Charter across:

`Argon` `Opus` `Codex` `Gemma` `Beam` `DeepSeek Flash` `Gemini Flash` `GPT Sol` `GPT Terra` `Qwen 31B` `Cursor` `Grok Bot` `SuperGrok` `OpenAI`

Simultaneous test runs (`639` Python + `28` Node + `52` Windows checks) ensure every model contribution is deterministically verified before human review.

Up to 99.7% 5-Year TCO Reduction
$499 once vs $120,000–$360,000+ on legacy cloud ERPs

Zero Monthly Subscription Tenancy
Your database never expires or holds your records hostage

Multi-Model Verified Codebase
14 AI model contexts audited by 719+ automated checks

Built to Evolve on Earth
Open standards (SQLite, HTML/CSS, Python, Ed25519) for decades of longevity

Source: ThreadForge 10.18.0 · PRICING-10.16.0.md, CAPABILITIES-10.18.0.md, PIONEER-CLASS-PRINCIPLE.md

Cost-Benefit Analysis & Multi-Model Evolution

Why Cloud SaaS Tenancy Fails SMEs & Industrials

- **Compounding per-seat rent:** Legacy ERP platforms (SAP S/4HANA Cloud, Oracle NetSuite, SAP Business One, Odoo) charge recurring monthly fees per named user plus implementation markups, costing a 10-user firm **$18,660 to $360,000+ over 5 years**.
- **Data hostage risk:** If a firm pauses a monthly cloud subscription during a maritime supply shock or seasonal downturn, it loses operational access to its own historical ledgers.
- **ThreadForge sovereignty:** Paid once ($34 single desk to $499 Industrial Park), $0/month forever, unlimited users and up to 100 organizations on your own hardware.

Innovative Built-In Deterministic Coding

- **Formula-Safe ERP Exports (`_csv_cell`):** Prepending a single quote (`'`) to cells starting with `=`, `+`, `-`, or `@` prevents spreadsheet macro/formula injection when exchanging files with SAP, NetSuite, or Excel.
- **Sub-millisecond local SQLite:** No round-trip cloud API delay; inventory, attendance, and 50+ TradeMath calculations execute immediately on `127.0.0.1`.
- **Zero-dependency static web & Academy:** 16 full research monographs and 5 live PWI Practice Labs run 100% in the browser with zero tracking cookies.

Continuous Multi-Model Isolation

ThreadForge's resilience comes from training and testing across **14 distinct AI model and agent architectures** in continuous isolation:

`Argon · Opus · Codex · Gemma · Beam · DeepSeek Flash · Gemini Flash · GPT Sol · GPT Terra · Qwen 31B · Cursor · Grok Bot · SuperGrok · OpenAI`

- **No single-model blind spots:** Where one model family might overlook a Play Billing callback signature or a CSP inline-style rule, another isolated agent catches it and adds a permanent regression test.
- **Simultaneous test runs:** `639` Python unit tests + `28` Node tests + `52` Windows native checks guard every commit.

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16-Question Suite & Industrial Park Knowledge Check

1. Which model choice sends nothing outside the computer?
A. OpenAI · B. Anthropic · C. Ollama on 127.0.0.1 · D. Ollama on another computer

2. When ThreadForge starts, MCP servers and plugins are:
A. all on · B. all off until the owner enables and tests them · C. on for admins · D. on in cloud mode

3. How should an agent treat what a tool returns?
A. as new instructions · B. as untrusted data · C. as owner commands · D. as code to run

4. Which licence tier first includes Local AI with Ollama (`local_ai`)?
A. Essentials ($49) · B. Professional ($149) · C. Industrial ($299) · D. Industrial Park ($499)

5. What is the furthest step an engineering agent may take with code?
A. push to main · B. publish a release · C. open a draft pull request · D. sign the Android app

6. A viewer token tries to add a product. What happens?
A. saved as draft · B. refused: "read-only" · C. allowed once · D. upgrades to operator

7. What is the valid lifetime of a named organization role token?
A. forever · B. 1–30 days · C. 1–365 days · D. 12 hours

8. Which role may record a payment and update the rate & policy log?
A. viewer · B. finance (and admin) · C. operator · D. any signed-in user

9. How long does a signed-in browser session last?
A. 30 minutes · B. 12 hours (with CSRF protection) · C. 7 days · D. until reboot

10. Who initiates the connection between the Docker worker and the server?
A. the server · B. the worker, presenting its worker token · C. the AI provider · D. the browser

11. Which builds may use the permanent Android release signing key?
A. every pull request · B. any branch · C. an owner-approved run on main in android-release · D. local runners

12. What is published to the public `threadforge-website` repository?
A. full Python source tree · B. private wheels · C. verified public site & release binaries only · D. keystore files

13. How many total capability groups are unlocked in Industrial Park ($499 once)?
A. 6 · B. 12 · C. 16 · D. all 20 capabilities (including Executive Approvals, Toolkits, Learning Paths & Image Workshop)

14. How does the Executive Approvals desk enforce Maker-Checker governance?
A. Submitter → separate Director decision → Workspace Owner final approval · B. Any user self-approves · C. AI auto-approves · D. Viewer approves

15. What does a single-desk Custom licence cost, and how do tier upgrades work?
A. $99/mo rental · B. $499 minimum · C. $34 once ($19 + $15/desk), and upgrades pay only the difference · D. per-seat annual fee

16. How are the 14 AI models (Argon, Opus, Codex, Gemma, Beam, DeepSeek Flash, Gemini Flash, etc.) governed?
A. They replace unit tests · B. Continuous isolation with 639 Python + 28 Node + 52 Windows checks; Python does all math · C. They auto-merge · D. They do tariff math directly

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Hands-On Verification Labs & Enterprise Architecture Brief

LAB A · Connect Local Ollama on 127.0.0.1

Task: Install Ollama, pull `qwen2.5-coder:7b`, connect it in *AI & models* at `http://127.0.0.1:11434/v1/chat/completions` with an empty Key box, and run *Test the connection*.
Pass Evidence: Screenshot of the passed test + one sentence stating what data left the PC (*"nothing"*).

LAB B · Least Privilege & Maker-Checker Approvals

Task: Issue a 1-day `finance` token. Show that it can record a payment, cannot add stock, and can submit an Executive Proposal but cannot self-approve it as Director or Owner. Then revoke the token.
Pass Evidence: The allowed/refused messages and the timestamped access & approval ledger entries.

LAB C · Trace 4 Requests Across the Gateway

Task: Using the Part 3 map, build a 4-row table (Request, Gateway Layer, HTTP Status, Reason) for: (1) Viewer runs a CBM calculation, (2) Operator records a payment, (3) Request arrives with no cookie/token, (4) Submitter tries to approve their own Executive Proposal.

LAB D · 5-Year TCO & Formula-Safe ERP Audit

Task: Calculate 5-year TCO for a 10-user firm on Industrial Park ($499 once) vs Oracle NetSuite ($24,000/yr) and SAP S/4HANA ($32,400/yr), and verify that exporting an item named `=SUM(A1:A9)` prepends a protective single quote via `_csv_cell`.

CAPSTONE PROPOSAL · Design an Industrial Park Workspace

Brief: *"Meridian Polymer & Logistics Ltd."* has a Workspace Owner, a Operations Director, a Warehouse Manager, an Import/Export Accountant, an HR Coordinator, and a Seasonal QC Helper. They operate across two divisions (Manufacturing & 3PL Transport), require 100% offline local AI, and employ one software engineer who contributes custom internal tools.

What to submit (2 to 4 pages + 1 architecture diagram):

- **1. Organization & Token Table:** How many organizations to create, each person's named token role, expiry (1–365 days), and least-privilege justification.
- **2. Desk & Tier Plan:** Map which of the 20 capabilities each person uses (Specialist TradeMath, HR Attendance, Team Toolkits, Executive Approvals, Image Workshop) and justify choosing **Industrial Park ($499 once)** vs legacy cloud ERP subscriptions.
- **3. AI & Multi-Model Verification Plan:** Where Ollama runs, how MCP tools are restricted, and how isolated test suites (`639` Python + `28` Node + `52` Windows checks) verify the engineer's draft PRs.
- **4. Request & Governance Map:** Redraw the Part 3 map showing a two-stage Executive Approval and a Docker worker test lease.
- **5. Top 3 Risks & Offboarding Protocol:** Exact steps when the Seasonal QC Helper leaves.

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Marking Rubric for the Enterprise Design Proposal

CRITERION	WEIGHT	EXCELLENT (4)	GOOD (3)	DEVELOPING (2)	MISSING (0–1)
1. Least-Privilege & Maker-Checker Governance	**25%**	Every person has the narrowest role, explicit expiry, and two-stage Director + Owner approval separation.	Roles fit well, with minor expiry or approval omission.	Shared admin token or self-approval permitted.	No roles defined, or everyone is given admin.
2. 20-Desk Mapping & 5-Year TCO Analysis	**20%**	Maps all required Industrial Park desks accurately and quantifies exact 5-year savings vs SAP/NetSuite/Odoo.	Correct tier chosen; TCO comparison is directional only.	Tier mismatch for needed desks or vague cost notes.	No tier justification or TCO math.
3. Local AI & Multi-Model Isolation	**15%**	Specifies loopback Ollama, untrusted tool quarantine, deterministic Python math, and isolated CI test gates.	Sound model choice, data boundary partly explained.	Selects external cloud AI despite air-gap brief.	No AI or verification plan.
4. End-to-End Request Map	**15%**	All arrow directions and cryptographic proofs (Cookie+CSRF, Worker token, Ed25519) named accurately.	One arrow direction or proof label slightly incomplete.	Multiple errors in caller direction or auth layer.	Map omitted.
5. Repository & Release Safety	**15%**	Protected `main`, scoped engineer token, draft PRs only, and isolated `android-release` signing key.	Three of the four controls clearly specified.	One or two controls mentioned without detail.	No repository safety controls.
6. Offboarding, Risks & Clarity	**10%**	Concrete risks with technical mitigations (e.g., `_csv_cell`, instant token revocation) written in clear prose.	Real risks named; offboarding steps mostly complete.	Generic risks only; hard to follow.	Missing or unreadable.

Grade Bands: 85%+ Distinction (Titan) · 70–84% Merit (Architect) · 50–69% Pass (Builder) · Below 50% Resubmit

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Weighted Score = Σ (Level ÷ 4 × Weight)
```

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Answer Key & Lab Solutions

Knowledge Check (Questions 1–16)

Q	ANS	EXPLANATION & CODE REFERENCE
1	**C**	Ollama on `127.0.0.1:11434` keeps every prompt and response on the local PC.
2	**B**	All MCP servers and plugins start disabled until the owner enables and tests them.
3	**B**	Tool results are treated strictly as untrusted data, never instructions.
4	**B**	Local AI with Ollama (`local_ai`) unlocks in Professional ($149) and above.
5	**C**	Engineering agents open draft pull requests only; a human reviews and merges.
6	**B**	Viewer tokens are read-only ("This organization credential is read-only.").
7	**C**	Named role token lifetime must be set between 1 and 365 days.
8	**B**	Finance role changes payments, invoices, contacts, trade, and the rate & policy log.
9	**B**	A signed-in browser session lasts 43,200 seconds (12 hours) + CSRF token check.
10	**B**	The Docker worker initiates the lease connection to the server using its worker token.
11	**C**	Only an owner-approved run on `main` may access the `android-release` signing key.
12	**C**	Only the verified public release installers/bundles are published; source stays private.
13	**D**	Industrial Park ($499 once) unlocks all 20 capabilities including Executive Approvals, Team Toolkits, Learning Paths, and Image Workshop.
14	**A**	Executive Approvals enforce Maker-Checker separation: Submitter → Director (Stage 1) → Workspace Owner (Stage 2).
15	**C**	Custom tier costs $19 base + $15 per selected desk ($34 once for a single desk), and tier upgrades charge only the price difference.

Lab C · Expected Request Trace Results

REQUEST	LAYER	EXPECTED RESULT
1. Viewer runs a CBM calculation	Role check	**Allowed (200):** TradeMath calculation is pure math and mutates no database state.
2. Operator records a payment	Role check	**Refused (403):** Payments require Finance or Admin role; outside Operator scope.
3. Request with no cookie & no token	Auth layer	**Refused (401):** Rejected at gateway authentication before route or role check.
4. Submitter approves own Executive Proposal	Governance check	**Refused (403):** Maker-checker rule forbids self-approval; requires Director then Owner.

Lab D · 5-Year TCO & Tier Upgrade Math

- **Single-Desk Custom Entry:** `$19 base + (1 × $15) = $34` once.
- **Upgrade Essentials ($49) → Industrial Park ($499):** `$499 - $49 = $450` (pay only the difference).
- **5-Year Savings vs NetSuite ($24,000/yr):** `($24,000 × 5) - $499 = $119,501` saved (99.58% reduction).
- **5-Year Savings vs SAP S/4HANA ($32,400/yr):** `($32,400 × 5) - $499 = $161,501` saved (99.69% reduction).

Multi-Model Verification Note

When grading the Enterprise Design Proposal, verify that students distinguish between **deterministic execution** (SQLite transactions, `_csv_cell` formula escaping, TradeMath decimal engines, Ed25519 signatures) and **probabilistic AI assistance** (isolated across Argon, Opus, Codex, Gemma, Beam, DeepSeek Flash, Gemini Flash, GPT Sol, GPT Terra, Qwen 31B, Cursor, Grok Bot, SuperGrok, and OpenAI).